

Connecting an SPE Expert amplifier to Decodium 4

Expert 1.3K-FA, 1.5K-FA and 2K-FA. Once connected, the DECØMETER shows power and SWR measured **at the amplifier** instead of at the exciter.

Before you start

The Expert amplifiers have two ports, **USB** and **RS-232**. They work independently but **never at the same time** — pick one and stay on it.

USB needs nothing special: a standard cable, and Windows presents the amplifier as a new COM port. RS-232 needs a cable wired as below, because the amplifier does not use the usual pins:

Amplifier DB-9/M		PC DB-9/F
pin 7 TX	----->	pin 2 RX
pin 8 RX	<-----	pin 3 TX
pin 5 GND	-----	pin 5 GND

Two requirements rule out some units: the firmware must be **15_05_15 or newer** for the protocol to exist at all, and on the 2K-FA the RS-232 port is only fitted from a certain serial number onward — on earlier units USB is the only way in.

The serial port problem, and how to solve it

Windows gives a serial port to one program at a time — reading included. If KTerm, the manufacturer's terminal, is open on that port, Decodium cannot have it. There is no way around this in Windows itself.

If you never need KTerm while operating, close it and skip this section.

If you want **both** — KTerm's full front-panel view *and* Decodium's readings — you need a **virtual serial port program** that duplicates one physical port into several virtual ones. Two that work:

VSPE (Virtual Serial Ports Emulator)

Free for 32-bit Windows, paid for 64-bit. Create a device of type **Splitter**: the source is the amplifier's real port, and it produces a virtual port that a second program can open. Point Decodium at the virtual one and leave KTerm on the real one.

com0com with hub4com

Free and open source. Note that `com0com` on its own does **not** share a physical port — it creates linked pairs of virtual ports. The piece that splits one real port among several programs is **hub4com**, which ships with the same project. Run it with the amplifier's real port as input and two virtual ports as outputs.

Whichever you use, Decodium goes on the copy, never on the real port. The physical port stays with the program doing the asking.

Setting it up

1. Connect and power on

The amplifier does not need to be in *Operate* — it answers in *Standby* too, which makes this easier to test off the air.

2. Press Search

In **Settings** → **Radio**, find the **AMPLIFIER** block and press **Search** next to *Amplifier port*.

Decodium asks every free serial port for a status report and watches for a reply in the SPE protocol. Whatever answers is the amplifier — it names its own model. On success the port, the speed and the reading mode are filled in for you, and the button reports what it found: `found 20K on COM7`.

The ports your CAT is using are left out of the search. Opening one, even just to ask a question, would take the radio away from whatever is driving it.

3. If the search finds nothing, set it by hand

Find the port in Windows Device Manager, then fill in the block:

SETTING	VALUE
Read amplifier	on — the master switch for the block
Amplifier port	COM7 — or whatever Windows assigned
Speed	115200 — the amplifier adapts down on its own
Listen only	off — unless you are using a splitter, see below

About "Listen only". With this on, Decodium opens the port read-only and never asks anything. The amplifier only speaks when spoken to, so you would wait forever. Turn it on **only** when KTerm is running on the split port and already doing the asking — then Decodium reads the same conversation without putting a single byte of its own on the line.

4. Point the DECØMETER at the amplifier

Open the DECØMETER and switch its source from **EXC** to **AMP**. Without this it keeps showing the exciter's power, however well the amplifier is answering.

If the amplifier goes quiet — powered off, cable pulled — the meter shows `AMP SILENT – SHOWING EXCITER` and falls back rather than freezing on a stale number.

What you get

READING	ON RECEIVE	ON TRANSMIT
Model	20K / 13K — announced by the amplifier itself	
Output power	0 W	measured watts
SWR at the antenna	—	measured
SWR before the tuner	—	measured
Supply voltage	0.0 V	48.0 V typical
Supply current	0.0 A	drawn amps
Heatsink temperature	always live	
Warnings and alarms	always live	

Every reply carries a checksum, and Decodium **discards any frame whose checksum does not match** rather than showing what it decoded. A false reading while you are running full power is worse than no reading at all.

When it does not work

Decodium writes an `[AMP]` line for each of these — open the diagnostic log and search for it.

STATUS	WHAT IT MEANS
no port	The port field is empty. Not an error — a setup you have not finished.
busy	Another program holds the port, almost always KTerm. Close it, or split the port.
silent	The port opened but nothing came back. Check <i>Listen only</i> is off, the amplifier is powered, and on RS-232 that the cable follows the pinout above.
no amplifier answered	The search ran and no port replied. The log lists which ports it tried and which it skipped.

How the link works

Not needed to use it — useful when something is odd and you are watching the line. The link runs at 8 data bits, 1 stop bit, no parity, no flow control, up to 115200 baud. Nothing is sent unprompted in either direction.

Decodium -> amplifier	55 55 55 01 90 90 three sync bytes, one byte follows, "get status", checksum
Amplifier -> Decodium	AA AA AA 43 ... three sync bytes, 67 characters of comma-separated values, two checksum bytes, CR LF

The reply is one line of 19 fields: model, standby or operate, receive or transmit, memory bank, input, band, antenna and tuner state, power level, then the measurements. Decodium reads the length the amplifier declares rather than assuming 67, so a firmware update that shifts the string by a character or two does not break the reading.

Protocol details follow the SPE Application Programmer's Guide, rev. 1.1 — Expert 1.3K-FA, 1.5K-FA and 2K-FA. SPE offers no support for third-party software; this document describes how Decodium uses the documented interface. · Decodium 4 FT2 — IU8LMC